

Disproof of Conjecture 00000001050: A Relative-Trace Surjectivity Criterion with Nontrivial Kernel

TLMC submission

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1 The conjecture (verbatim)

“Conjecture: For any composition of a permutation polynomial f with a linearized polynomial, the image under the relative trace F_{q^2}/F_q is everything if and only if the kernel is trivial (a relative-trace surjectivity criterion).”

2 Verdict

FALSE. The biconditional fails already at $q = 2$ with the identity permutation polynomial: the relative-trace image is everything while the kernel is nontrivial.

3 Counterexample

Let $q = 2$ and $\mathbb{F}_4 = \mathbb{F}_2(\omega)$ with $\omega^2 = \omega + 1$, $\omega^3 = 1$. Take

$f = \text{id} \in \mathbb{F}_4[x]$ (a permutation polynomial), $L(x) = \omega x + \omega x^2 = \omega(x + x^2) \in \mathbb{F}_4[x]$,

where L is a linearized polynomial (exponents $2^0, 2^1$; $\mathbb{F}_2$ -linear map). The composition is $g = L \circ \text{id} = L$; since $f = \text{id}$, both composition orders coincide. The relative trace is $\text{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(z) = z + z^2$.

3.1 Pointwise computation

x	$x + x^2$	$L(x) = \omega(x + x^2)$	$\text{Tr}(L(x)) = L(x) + L(x)^2$
0	0	0	0
1	$1 + 1 = 0$	0	0
ω	$\omega + \omega^2 = 1$	ω	$\omega + \omega^2 = 1$
ω^2	$\omega^2 + \omega^4 = 1$	ω	$\omega + \omega^2 = 1$

3.2 The two halves of the biconditional

- **Kernel nontrivial.** $\ker L = \{x \in \mathbb{F}_4 : x + x^2 = 0\} = \{0, 1\} = \mathbb{F}_2 \neq \{0\}$.
(Indeed $x^2 = x$ has exactly the roots 0, 1 in any field.)
- **Trace image is everything.** $L(\mathbb{F}_4) = \{0, \omega\}$, so

$$\mathrm{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(L(\mathbb{F}_4)) = \{\mathrm{Tr}(0), \mathrm{Tr}(\omega)\} = \{0, \omega + \omega^2\} = \{0, 1\} = \mathbb{F}_2.$$

Hence “image is everything” holds while “kernel is trivial” fails, so the stated criterion $(\mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(g(\mathbb{F}_{q^2})) = \mathbb{F}_q) \iff (\ker g = \{0\})$ is false. Conjecture 00000001050 is disproven.

4 Lean verification

The file `lean4/Main.lean` encodes $\mathbb{F}_4$ as a four-constructor inductive type with explicit addition/multiplication/Frobenius tables and proves: `surjective_true` ($\mathrm{Tr} \circ L$ hits 0 and 1), `kernel_not_trivial` ($1 \in \ker L, 1 \neq 0$), and `criterion_false` ($\neg(\mathrm{Surjective} \leftrightarrow \mathrm{KernelTrivial})$). Every assertion is closed by `rfl` or constructor exhaustion; there are zero axioms and no `sorry`. `lean4/Check.lean` prints `#print axioms` for every theorem. An independent recomputation is available in `reproduce.py`.

5 Boundary

This counterexample only refutes the verbatim biconditional at the smallest instance ($q = 2, f = \mathrm{id}$). No claim is made about the converse direction (kernel trivial $\Rightarrow$ surjective) in general, nor about compositions with $f \neq \mathrm{id}$; both composition orders coincide here because $f = \mathrm{id}$.